



A Spot Pattern is Like a Fingerprint

Jaguar Identification Project

Who Am I?



■ Antonio Rueda-Toicen

- Computer Scientist
- AI Engineer and DevRel @ Voxel51
- Research Scientist at the Hasso Plattner Institute
- Instructor at NVIDIA's Deep Learning Institute and Berlin's Data Science Retreat



Team



Davide Panza



Shahabeddin Dayani



Aleksandra Kudaeva



Gerard de Melo



Abigail Martin

Agenda

- > An intro to the Jaguar Identification Project
- > An overview to image embeddings and dataset curation
- > Our upcoming Kaggle competition

PROTECT OUR JAGUARS

The Jaguar Identification Project is dedicated to the long-term study and conservation of jaguars in Brazil's Pantanal.

[Submit your photos](#)

[More about the jaguar](#)



What this project is about

- Conservation
- Education (both of biology and computer vision)
- Research on wildlife re-identification methods

[Jaguar Identification Project](#)



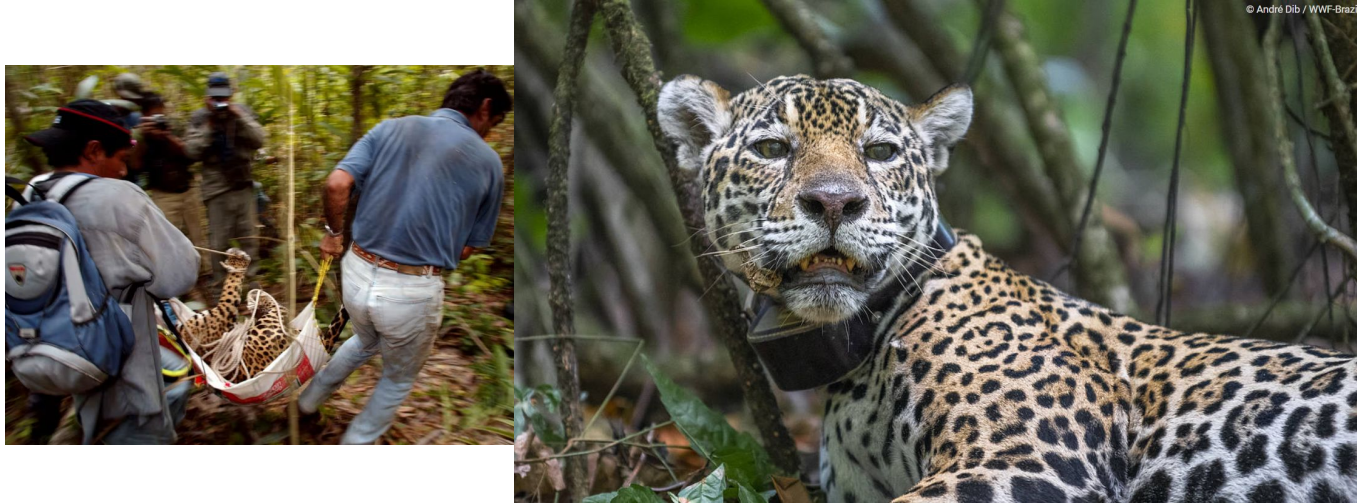
Protecting Endangered Species Starts with Understanding Their Threats

Wildfires
Deforestation
Illegal trade
Climate change
Human-wildlife conflicts



What Gets Measured, Gets Managed

Tracking animal population is a necessary but tedious task.



Extractional methods are invasive, expensive and dangerous

[WWF's first jaguar collaring mission reveals flourishing population on Amazonian island | WWF](#)

[Conservationists track jaguars from space](#)

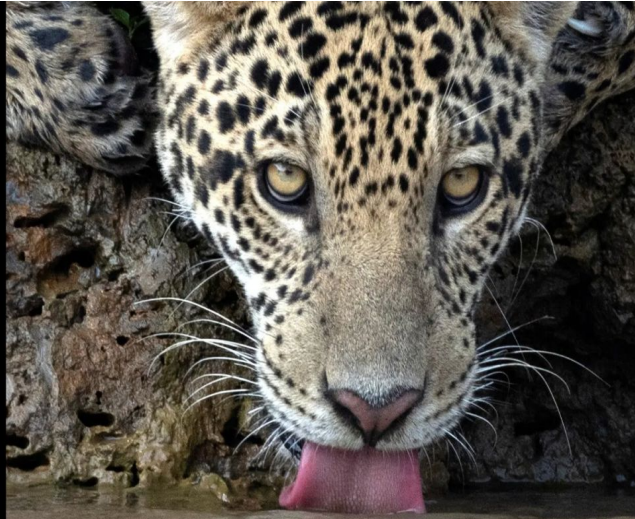
Jaguars in Porto Jofre

- Houses one of the largest wild jaguar populations
- The Pantanal National Park has 1,200 wild animal species
- Popular destination for ecotourism and wildlife observation



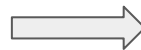
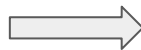
In the Land of the Jaguars

Brazil's Pantanal, the world's largest wetlands, is home to a sizable population of jaguars. Near one tiny outpost, the normally shy animals are remarkably easy to observe.



Tracking the Jaguar Population

Photos from eco-tourists and citizen scientists aid in tracking jaguar populations



Jaguar ID?
Medrosa
Patricia
Ousado
...

Solution: Develop a jaguar identification system capable of recognizing a specific jaguar's ID from a photo (e.g. we want to distinguish Medrosa from Patricia) for demographics

Identifying a jaguar



Right Side

Date: 26 July 2019

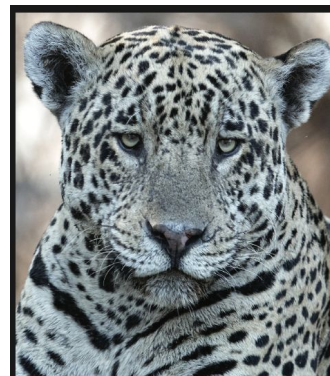
Photo Credit: Paul Donahue



Left Side

Date: Unknown

Photo Credit: Rodrigo Chávez Noriega



Date:

16 August 2020

Photo Credit:

Chris Brunskill

AJU

Pronunciation: Ah-joo

Other Names: Henry II

Name Translation: Mature in Tupi-Guarani language

Sex: Male

Seasons Observed:
2017, 2018, 2019, 2020, 2021,
2022, 2023

Relationships:
Cub of Courtney (2016), Mate of Estela (2018 - a day after she was seen with her cubs), Mate of Borboletinha (2019), Mate of Águe (2019), Mate of Estela (2020), Mate of Medrosa (2022), Mate of Pyte (2023)

Observations:
Captured on 9/19/20 - 110kg



+ add stage



Explore Enterprise



● Unsaved view

FILTER  

TAGS

☐ sample tags

☐ label tags

LABELS

1

1 ✓

☒ ground_truth

90 ^

confidence

Filter by confidence

id

+ filter by id

label

+ filter by label

☒ Katniss

 Show samples with label

Reset

tags

+ filter by tags

METADATA

2 ✓

☐ metadata.size_bytes

☐ metadata.mime_type

Samples

+

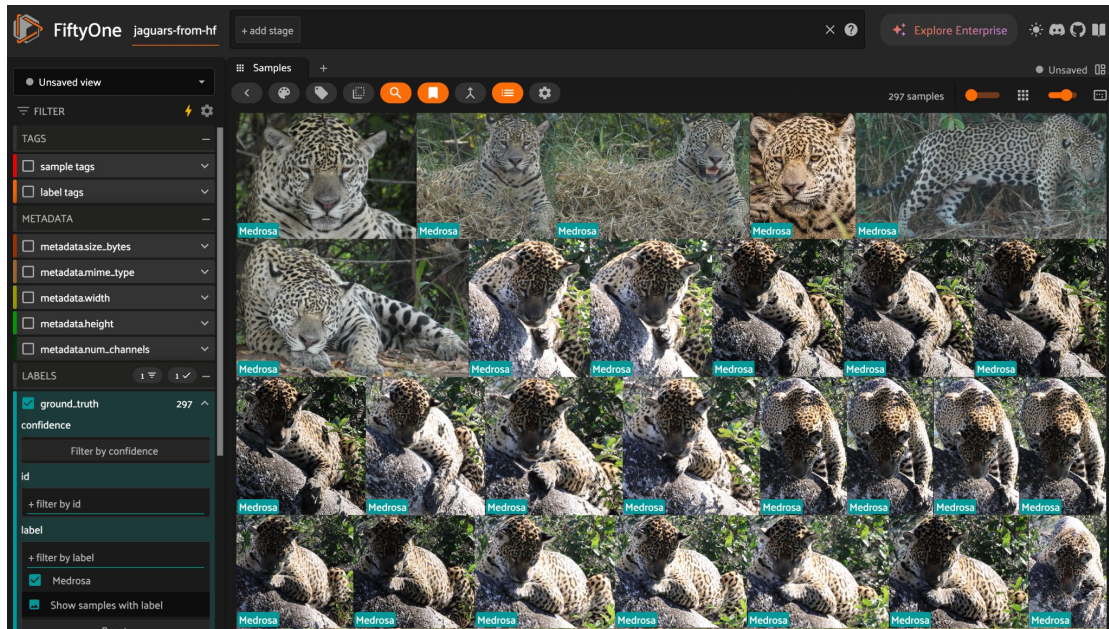
● Unsaved 



90 samples

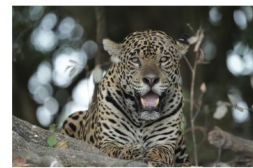


Digesting the jaguar field guide



INDIVIDUAL JAGUARS

32 Individual Jaguars: Males



Xingu - Credit: Thomas Schuler

52 Individual Jaguars: Females



Medrosa - Credit: Kike Arnaiz

33 Aju	43 Krishna
34 Bagua	44 Kwang
35 Bernard	45 Manath
36 Bororo	46 Merlin
37 Chevo	47 Odin
38 Donal	48 Ousado
39 Forasteiro	49 Rio
40 Inka	50 Tusk
41 Ipepo	51 Xingu
42 Juru	
53 Alira	65 Marcela
54 Axa	66 Margo
55 Benita	67 Medrosa
56 Borboletinha	68 Overa
57 Carvarinha	69 Oxum
58 Courtney	70 Patricia
59 Estela	71 Pixana
60 Guaraci	72 Pollyanna
61 Ibaca	73 Pyte
62 Jaju	74 Saseka
63 Jeni	75 Solar
64 Katniss	76 Ti

Overview > Core Capabilities > Curate

- > *Find*: filter, match, sort, select
- > *Remove*: duplicates
- > *Add*: tags, metadata, predictions
- > *Correct*: annotation mistakes



Split (4)

cropped_body · 3.1k rows

Search this dataset

filename	label	image
string · lengths	string · classes	image · width (px)
99	32 values	3086.93k
00000.jpg	Ousado	
00001.jpg	Ousado	
00002.jpg	Ousado	
00003.jpg	Ousado	
00004.jpg	Ousado	
00005.jpg	Ousado	
00006.jpg	Ousado	
00007.jpg	Ousado	

< Previous

123...31

Next >

Ask AI to help write your query...

-- The SQL console is powered by DuckDB WASM and runs entirely
-- Get started by typing a query or selecting a view from the
SELECT * FROM cropped_body LIMIT 10;

Subsets and Splits 1/4

cropped_bodyraw_images↓cropped_head↓segmented_body↓

Run Query

Saved QueriesTop Community Queries

No saved queries yet

Save your SQL queries to embed, download, and access them later. Queries will appear here once saved.



● Unsaved view

FILTER



TAGS

☐ sample tags☐ label tags

METADATA

☐ metadata.size_bytes☐ metadata.mime_type☐ metadata.width☐ metadata.height☐ metadata.num_channels

LABELS

2 ✓

☒ ground_truth☐ prediction☐ segmentations_body☐ bboxes_head☒ segmentations_head

PRIMITIVES

☐ id☐ filepath☐ created_at

Color settings

+



4,300 samples

Sort by



Ousado



Ousado



Ousado



Ousado



Ousado



Ousado



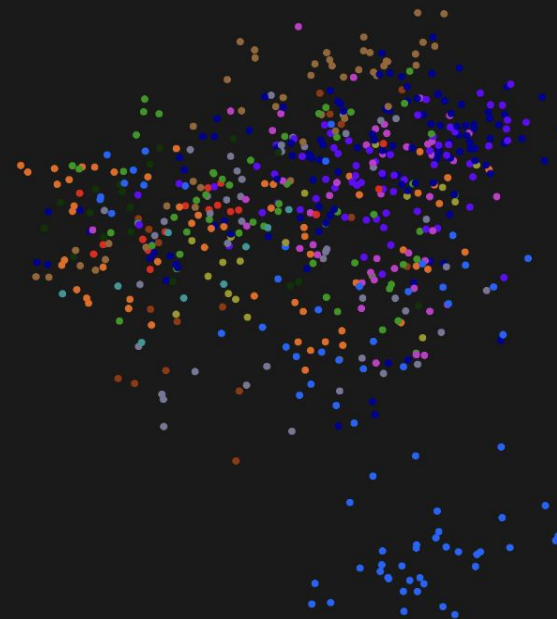
Embeddings

+

pca_dino_v2

ground_truth.label

+



● Unsaved

Abril

Aue

Akalo

Alira

Apeiara

Ariely

Bagua

Benita

Bernard

Borbotetinha

Bororo

Capi

Donal

Guaraci

Hero

Inka

Ipepo

Jaju

Juru

Kamaikua

Katniss

Kwang

Kyyavera

Lua

Luna

Madalena

Marcela

Medrosa

Ousado

Overa

Oxum

Patricia

Pixana

Pollyanna

Pyte



Unsaved view

FILTER

TAGS

☐ sample tags

☐ label tags

METADATA

☐ metadata.size_bytes

☐ metadata.mime_type

☐ metadata.width

☐ metadata.height

☐ metadata.num_channels

LABELS 2 ✓

☒ ground_truth

☒ uniqueness

☐ representativeness

PRIMITIVES

☐ id

☐ filepath

☐ created_at

☐ last_modified_at

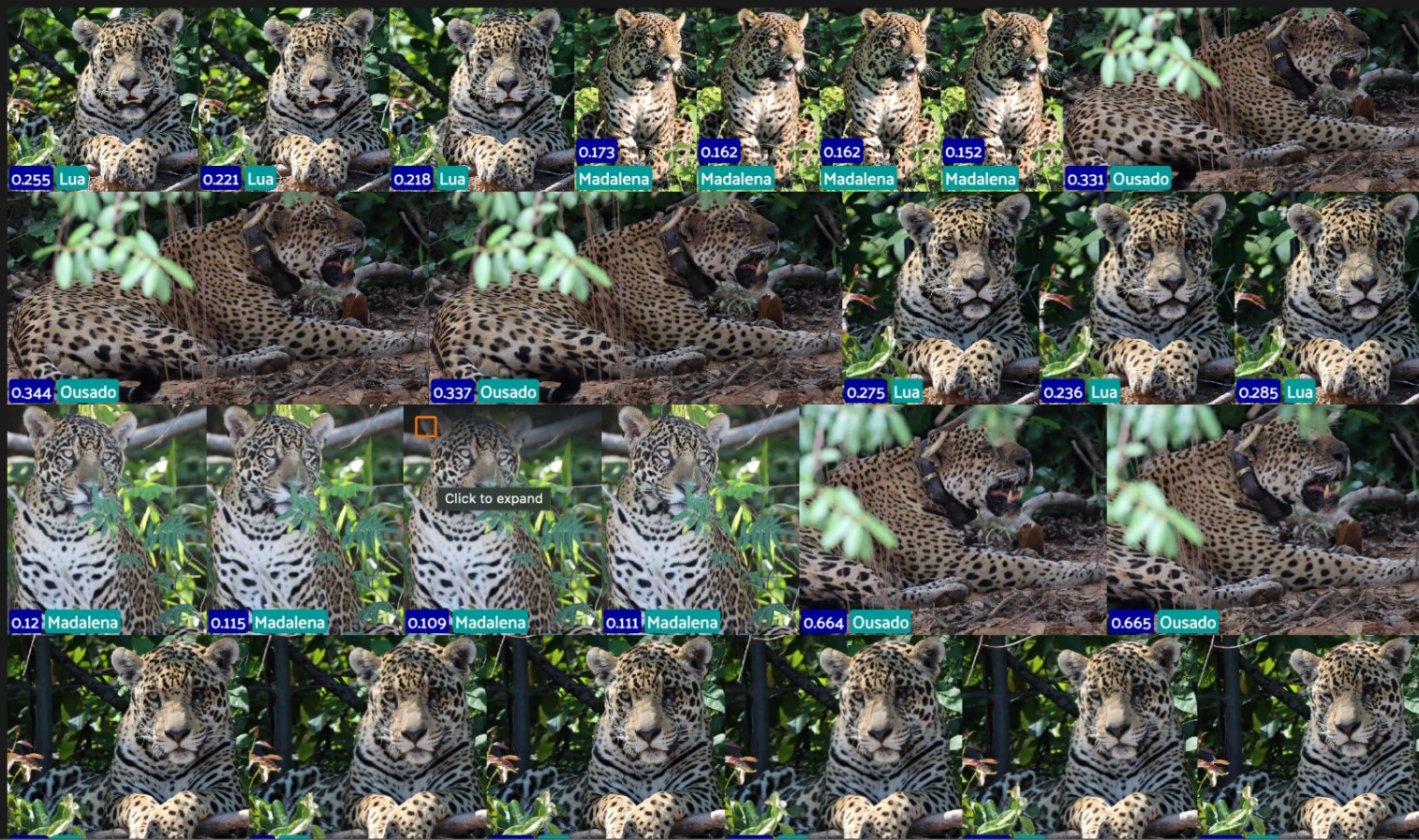
OTHER

Samples



2,935 samples

Sort by



Unsaved view

FILTER

- sample tags
- label tags

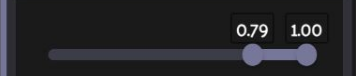
METADATA

- metadata.size_bytes
- metadata.mime_type
- metadata.width
- metadata.height
- metadata.num_channels

LABELS

ground_truth

uniqueness



Show samples with uniqueness

Reset

representativeness



Show samples with representativ...

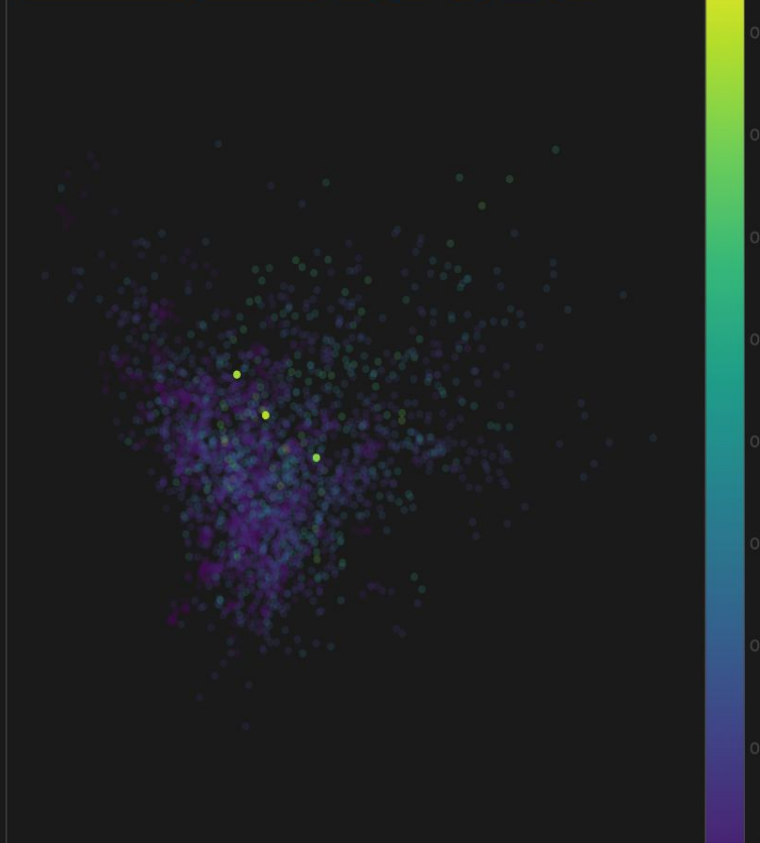
Samples

Navigation icons: back, palette, tag, grid, dropdown, 3 samples, zoom slider, grid icon, zoom slider, info icon.



Embeddings

mobilenet_pca uniqueness + icons



Unsaved view

FILTER

- ☐ sample tags
- ☐ label tags

METADATA

- ☐ metadata.size_bytes
- ☐ metadata.mime_type
- ☐ metadata.width
- ☐ metadata.height
- ☐ metadata.num_channels

LABELS 2 3

☒ ground_truth

☒ uniqueness 3

0.79 1.00

Show samples with uniqueness

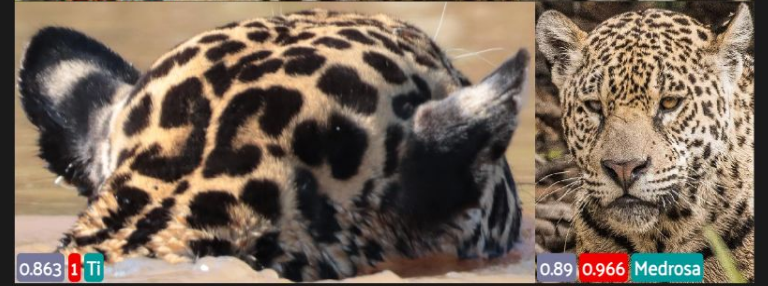
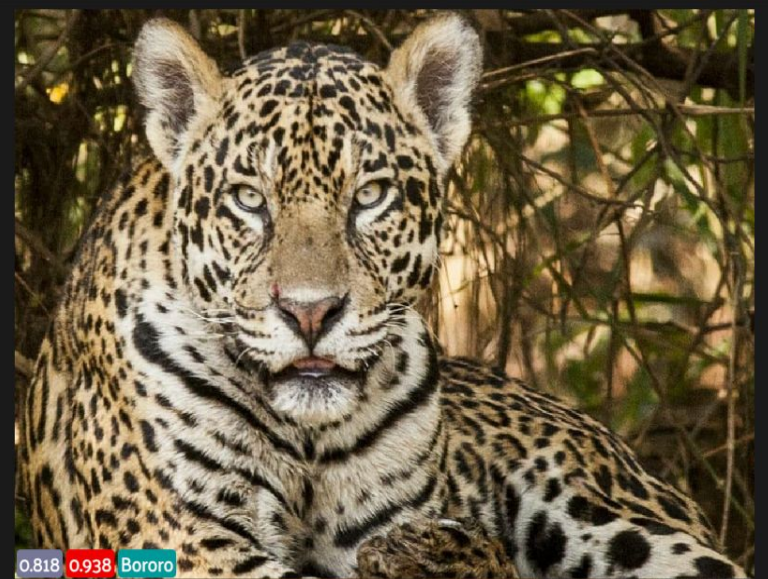
Reset

☒ representativeness 3

0.76 1.00

Show samples with representativ...

Samples + 3 samples



Embeddings x + Unsaved

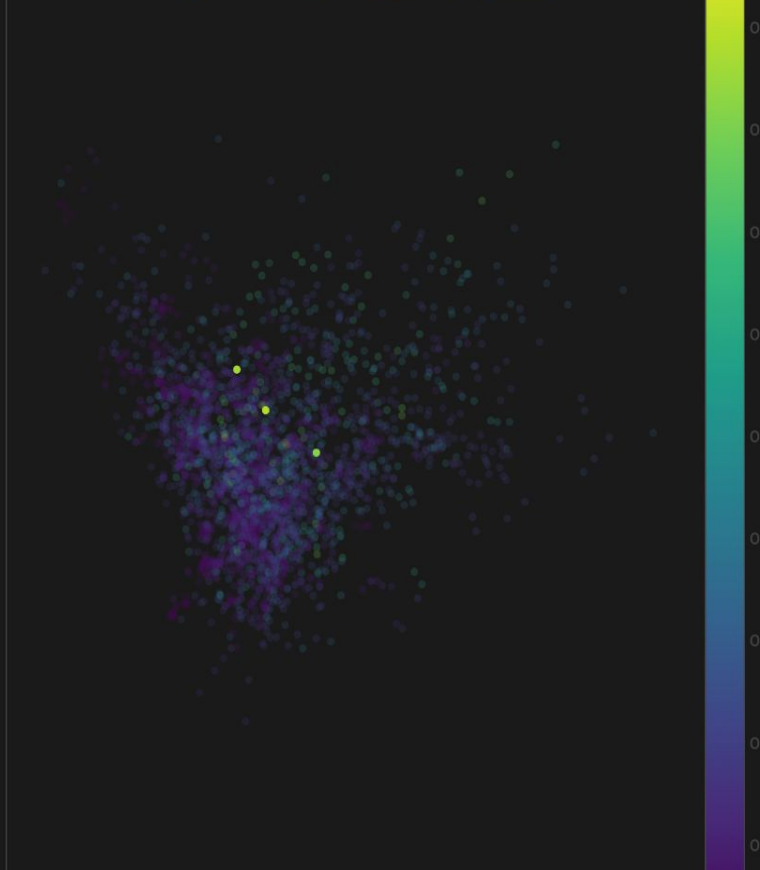
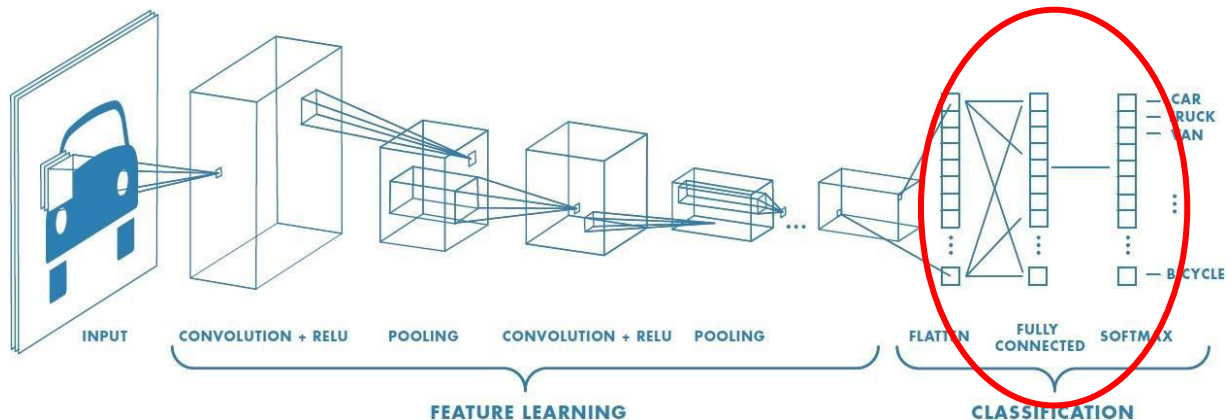


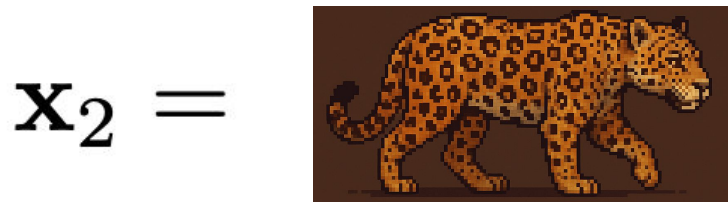
Image embeddings from neural networks

When we train a neural network, we make it **embed** inputs that are semantically similar, close to each other



Dense vector embeddings can be produced by flattening, pooling, or feeding (learned) features to linear layers

Quick intro to dense vector embeddings

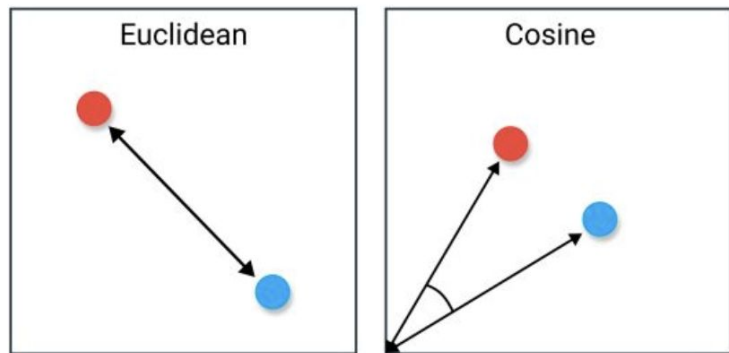


$$\mathbf{x}_1\text{sparse} = \mathbf{x}_2\text{sparse} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

$$\mathbf{x}_2\text{dense} = \begin{bmatrix} 0.10 & -0.75 & 0.35 & 0.41 & -0.20 & 0.89 & -0.60 & 0.03 & 0.59 & -0.50 \end{bmatrix}^T$$

$$\mathbf{x}_1\text{dense} = \begin{bmatrix} 0.12 & -0.85 & 0.37 & 0.44 & -0.22 & 0.90 & -0.63 & 0.01 & 0.58 & -0.49 \end{bmatrix}^T$$

Metrics to compare embeddings



$$d_{\text{euclid}}(x_1, x_2) = \sqrt{\sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

$$\text{cosine similarity}(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \|x_2\|}$$

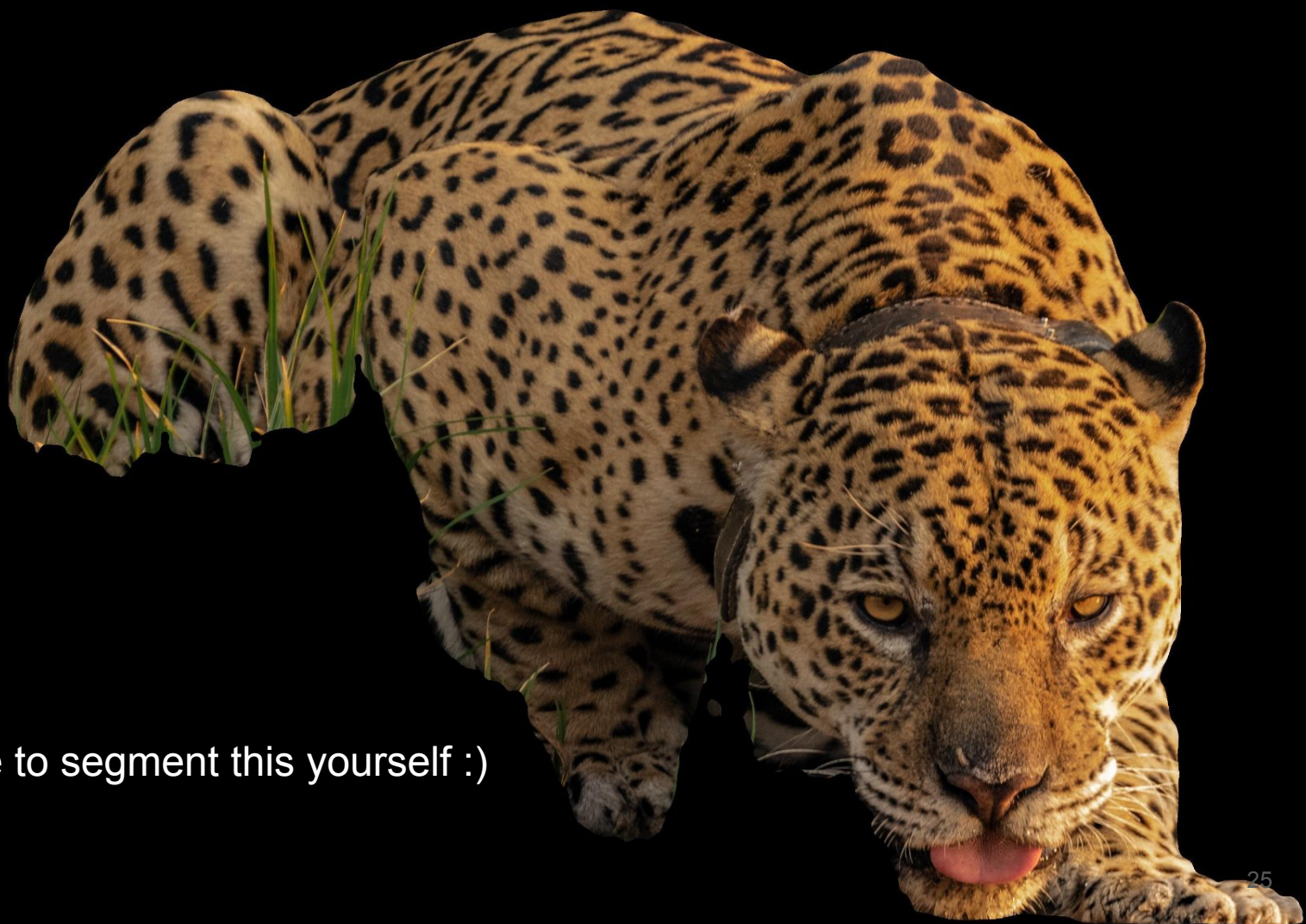
Upcoming Kaggle Competition

- Contribute to citizen science efforts at the Pantanal Jaguar ID Project
- Push the state of the art to identify wild jaguars from photos
 - Hypothesis : well curated images of the subjects avoid spurious correlations





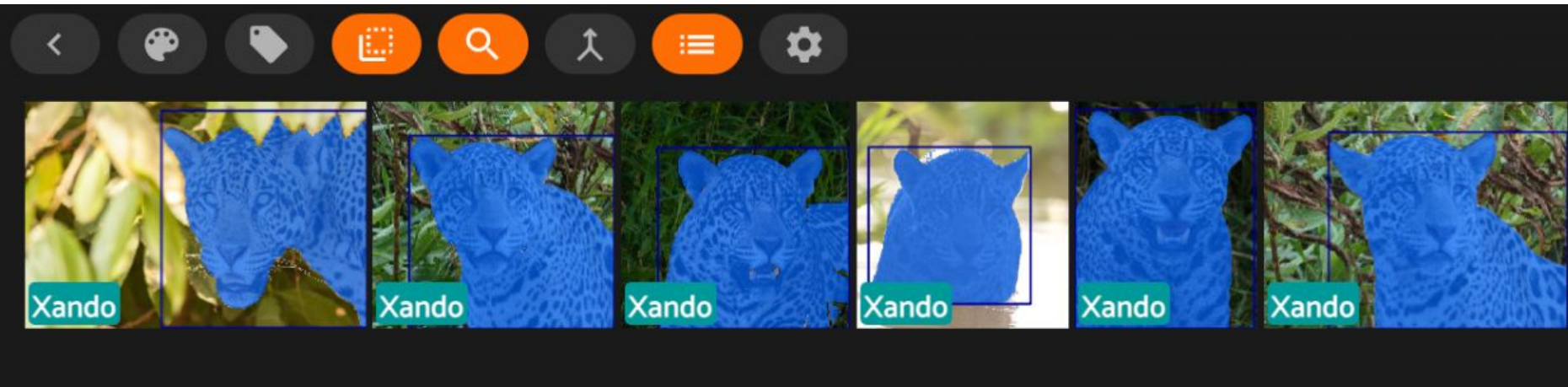
Kaggle
Test Set
Rule #1:
No
Background



Yes, you will have to segment this yourself :)



Open vocabulary detection and segmentation with YOLOE



```
import fiftyone as fo
from ultralytics import YOLOE

model = YOLOE("yoloe-11s-seg.pt")

names = ["body of a jaguar", "head of a jaguar"]
model.set_classes(names, model.get_text_pe(names))

dataset.apply_model(model, label_field="yoloe_segmentation")
```




+ add stage



Explore Enterprise



Unsaved view

FILTER



☐ metadata.size_bytes

☐ metadata.mime_type

☐ metadata.width

☐ metadata.height

☐ metadata.num_channels

LABELS 1 2

☒ ground_truth

☐ uniqueness

☐ representativeness

☒ yoloe_segmentation 743

confidence

0.25 0.50

Select detections with confidence

Reset

id

+ filter by id

index

Filter by index

label

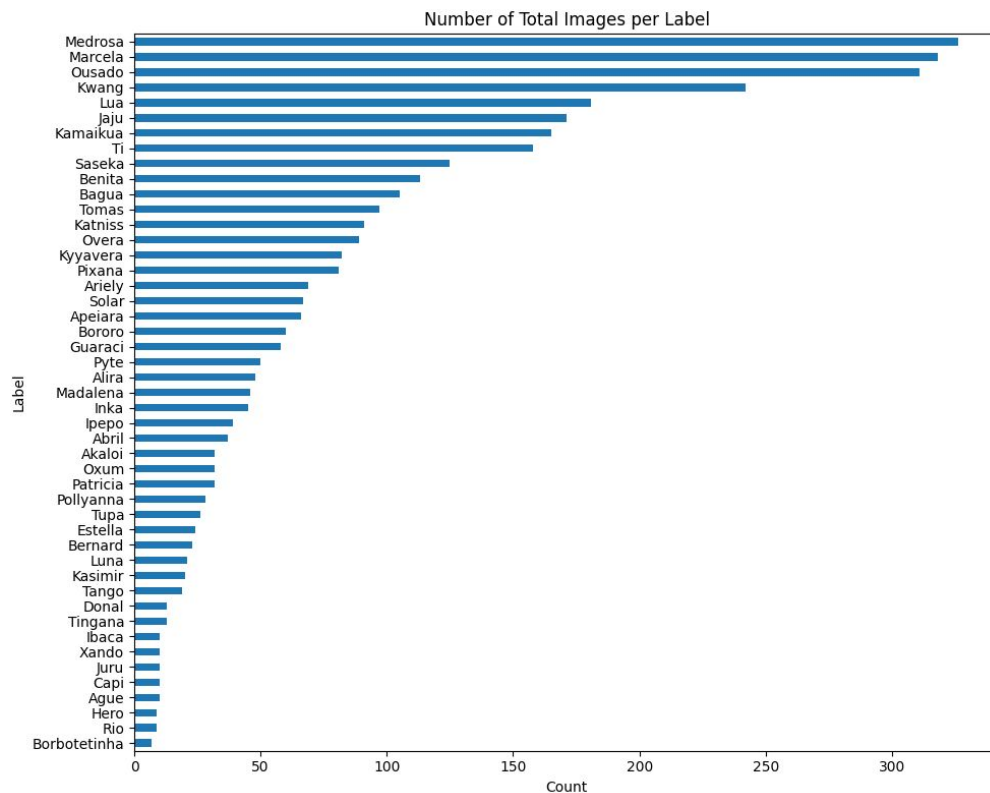
Samples



664 samples



Statistics of the images



Ousado

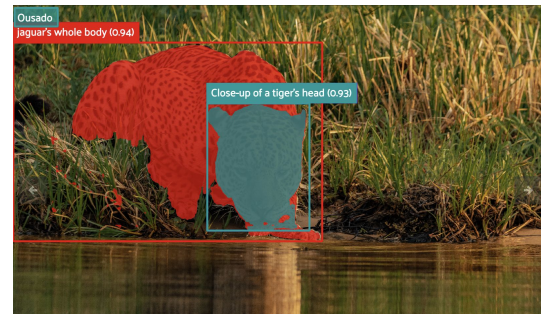
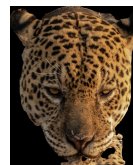
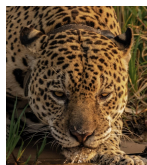
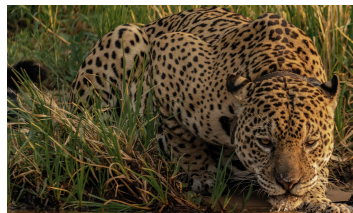
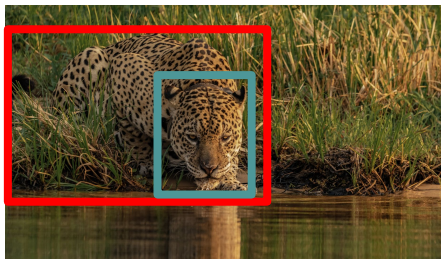
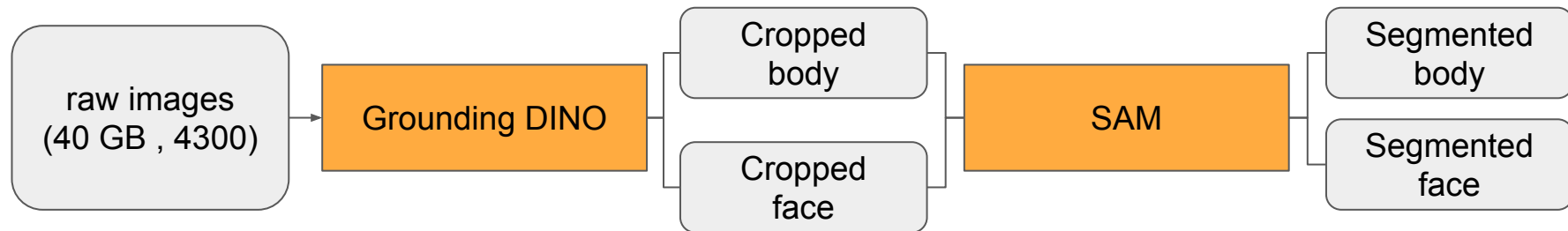


Medrosa

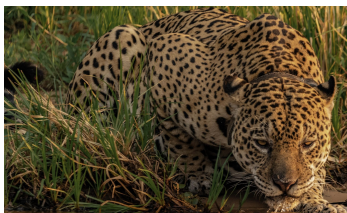
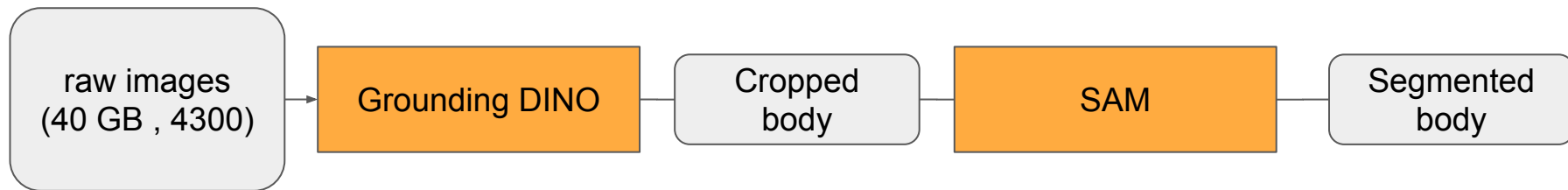


Patricia

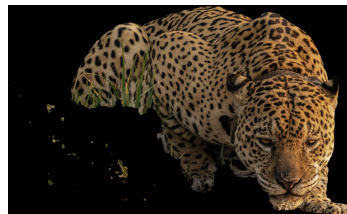
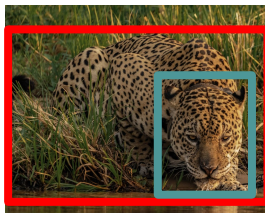
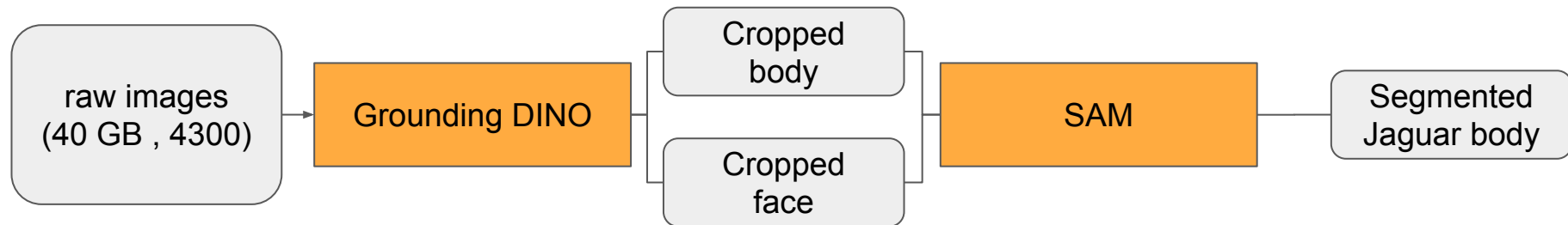
Dataset preparation pipeline



Dataset preparation pipeline

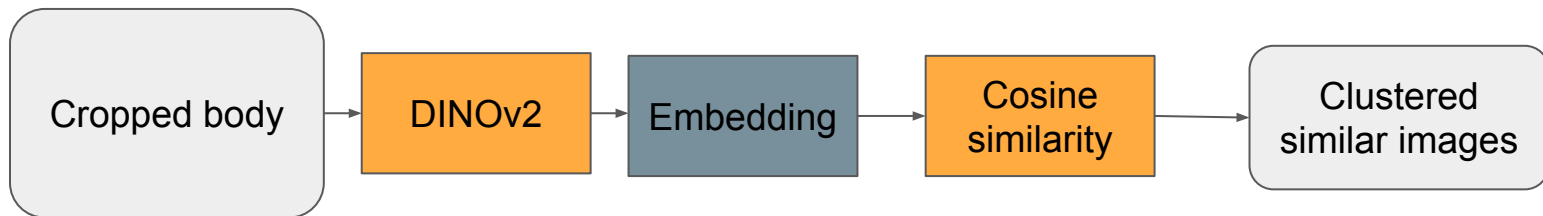


Dataset preparation pipeline

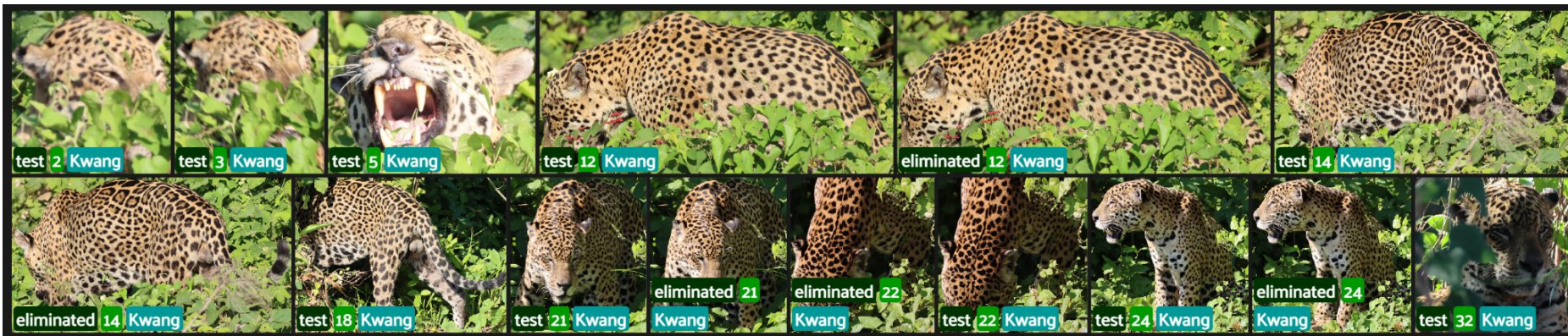
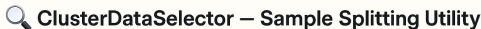


Issue: Near Duplicate Images

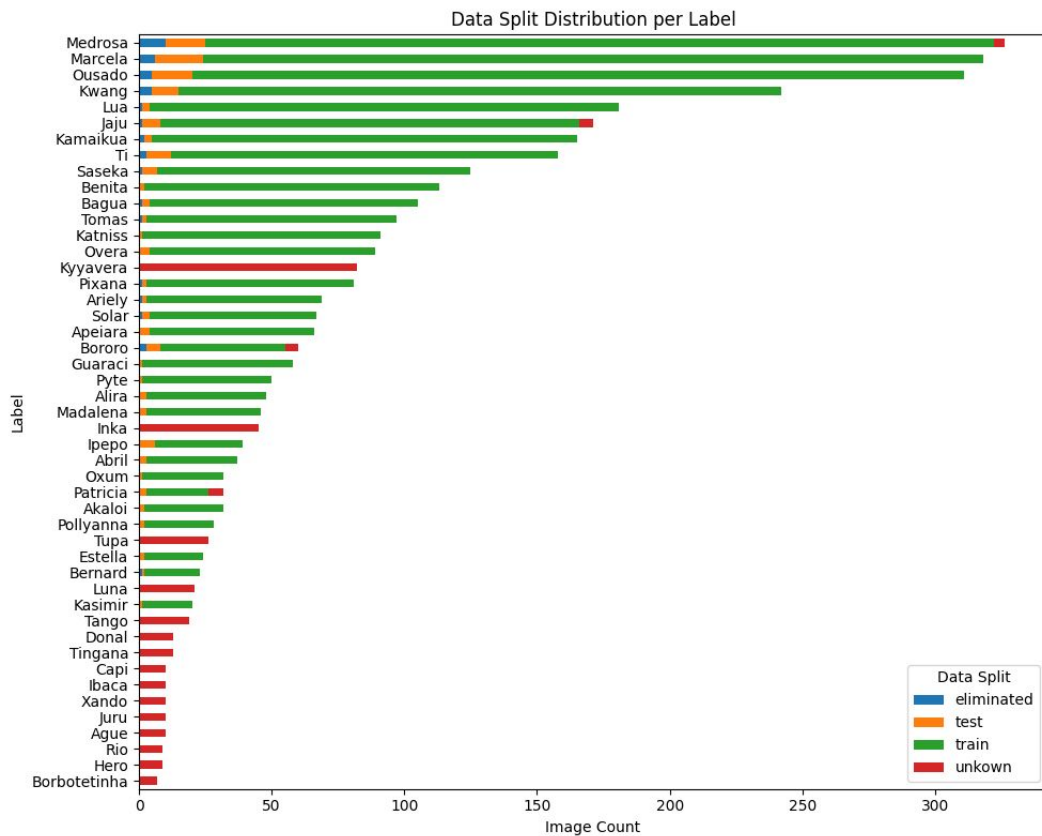
Leaky splits create a false and inflated performance benchmark



Deduplicated test / train split



Class balance after deduplication

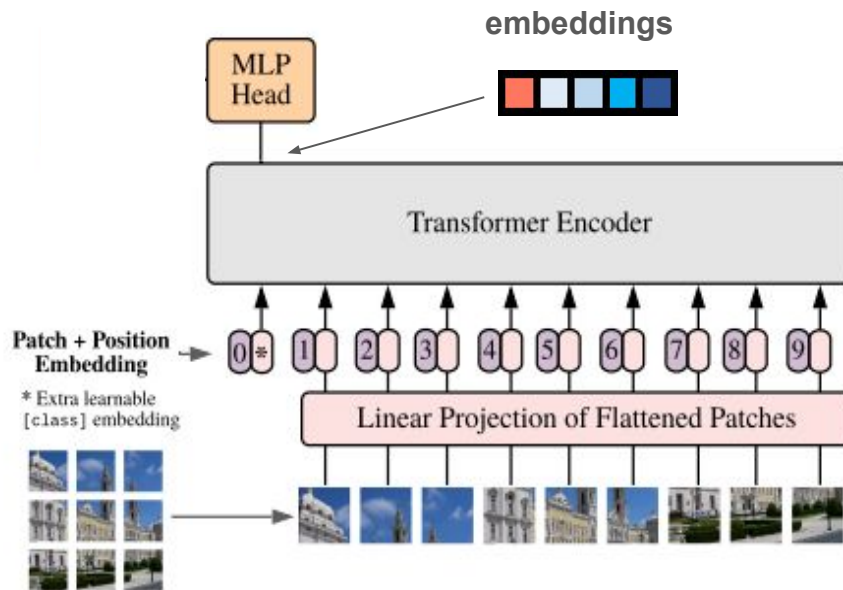


Jaguar-ID Classifier

1) Fine-tune a “backbone model” to get images embedding

Backbone: **DINOv2**

- A self-supervised vision transformer (encoder)
- Dense image embeddings
- Captures rich visual and semantic information

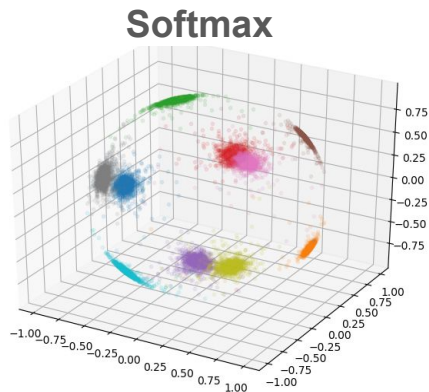
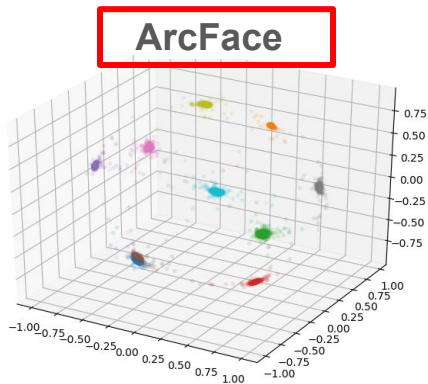


Jaguar-ID Classifier

1) Fine-tune a “backbone model” to get images embedding

Fine-Tuning Methods:

- [ArcFace loss](#) (variant of softmax)
- It forces images from the same jaguar to **cluster together**

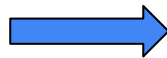
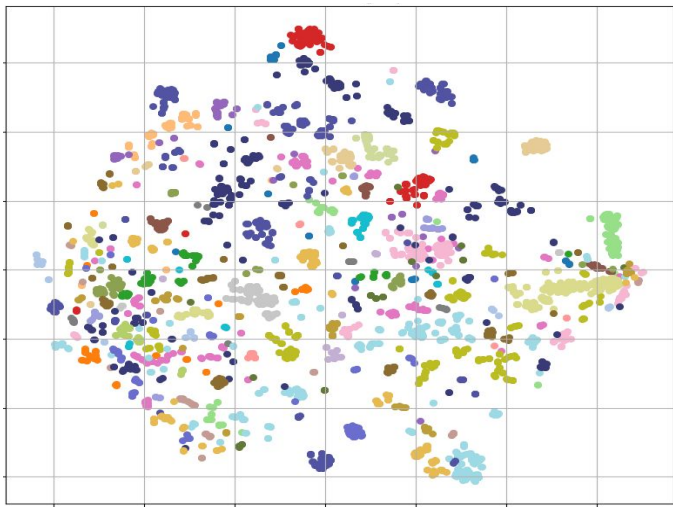


Jaguar-ID Classifier

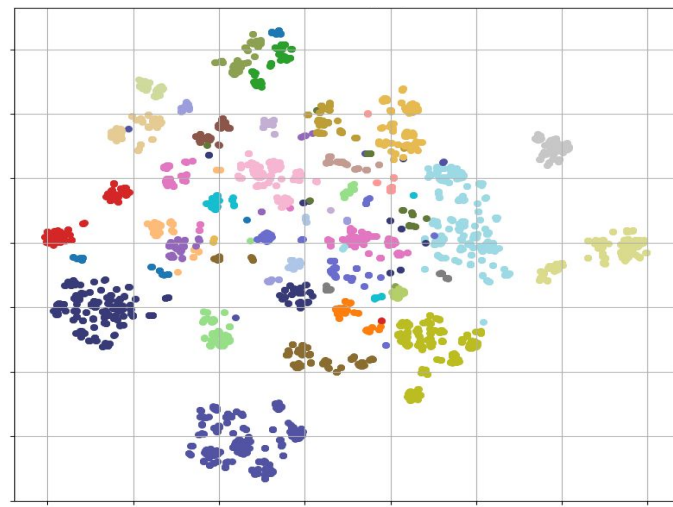
1) Fine-tune a “backbone model” to get images embedding



Pre Fine Tuning

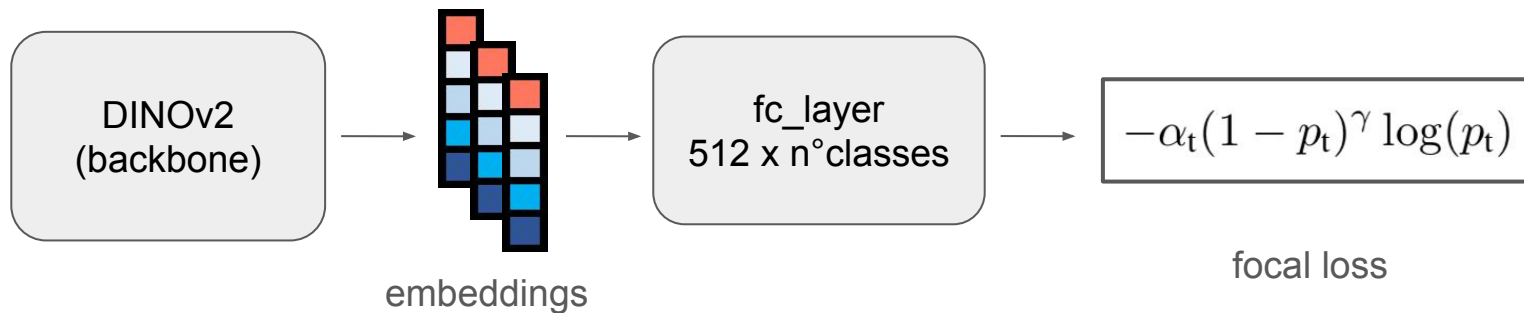


After Fine Tuning



Jaguar-ID Classifier

2) Add a classification layer to the backbone



Fine-Tuning Details:

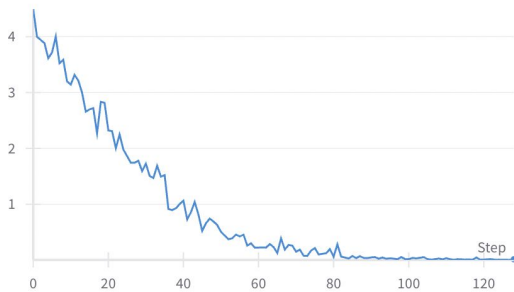
- Focal loss with weighted cross-entropy (because of unbalanced data)
- Data augmentation
- 8 epochs, LR: 3e-4, batchsize:128

Jaguar-ID Classifier

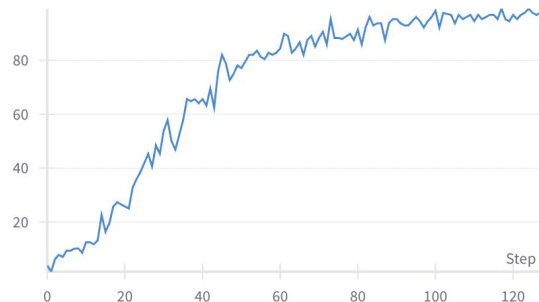
*mean reciprocal rank or f1 score are better metrics than accuracy

2) Add a classification layer to the DINOv2 backbone

Batch Loss

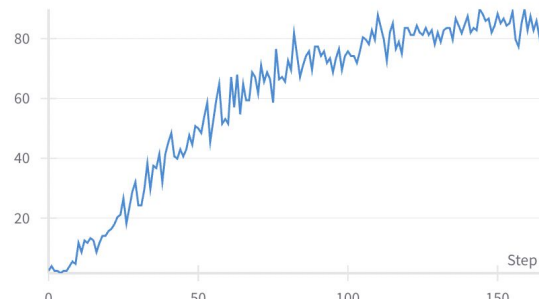
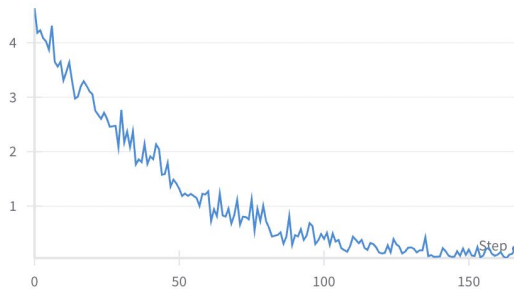
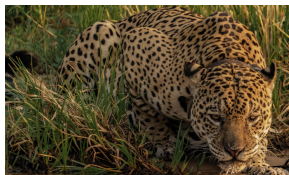


Batch Accuracy



Test Accuracy

~70%

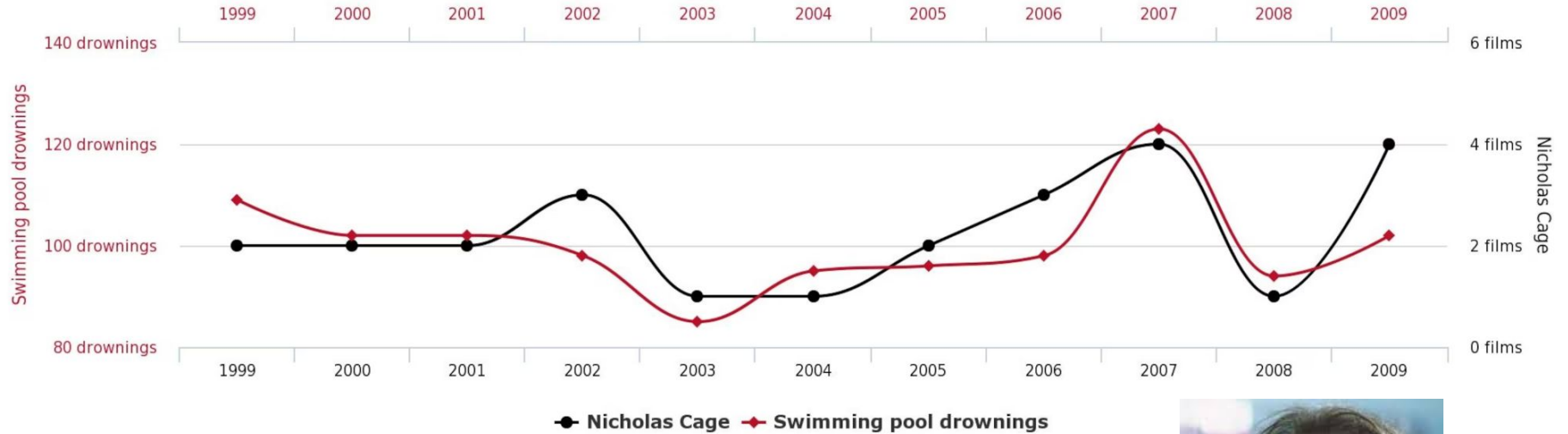


~62%

Number of people who drowned by falling into a pool

correlates with

Films Nicolas Cage appeared in



[Spurious Correlations in Machine Learning: A Survey](#)



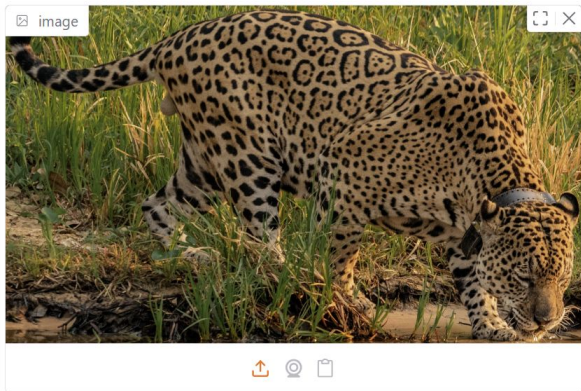
tylervigen.com

Gradio App on Hugging Face

Spaces shahabdaiani/jaguar_identification_app like 0 Running

Jaguar Identifier (DINOv2 + ArcFace)

Upload an image of a jaguar. The model will classify it among 33 known individuals.



Share via Link

Clear

Submit

[Jaguar Identification App - a Hugging Face Space](#)

Next challenge: camera traps



Resources

- [Jaguar Identification Dataset on Hugging Face](#)
- [Notebook to perform embeddings-based exploration and segmentation on FiftyOne](#)
- [Starter notebooks for dataset curation, classification and DinoV2 fine-tuning](#)
- [FiftyOne's Discord](#)
 - **#practical-computer-vision-workshops channel**



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